

Master Thesis
BlueRov2 Formation Control

Week 8

João Zenário

Plan for this week

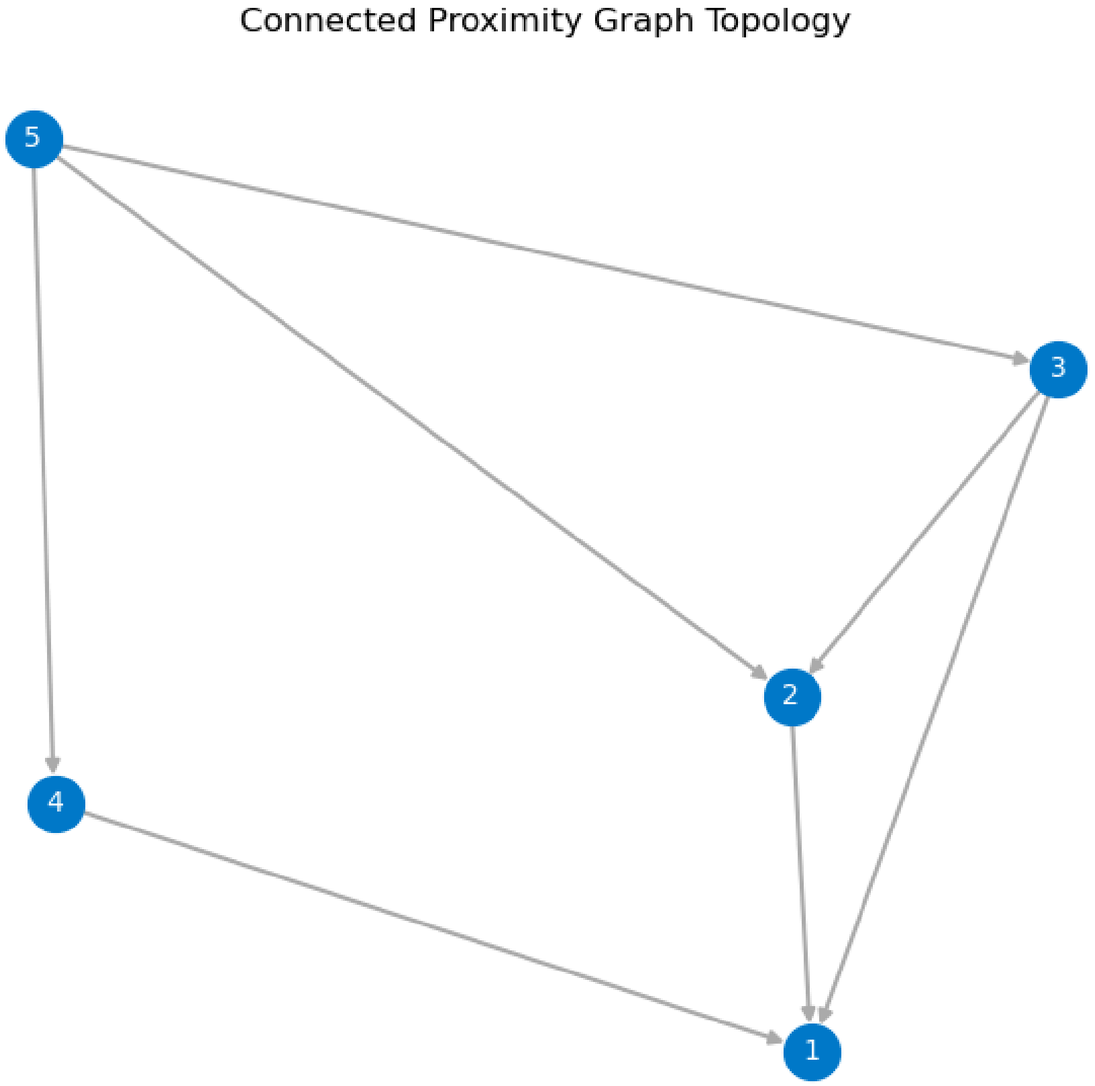
- Formation control with single integrator agents no constraints
  - Undirected graph
  - Directed graph
- Single integrator formation control with constraints
  - Only connectivity maintenance
  - Then also colision avoidance
  - Field of view
- Double integrator agents

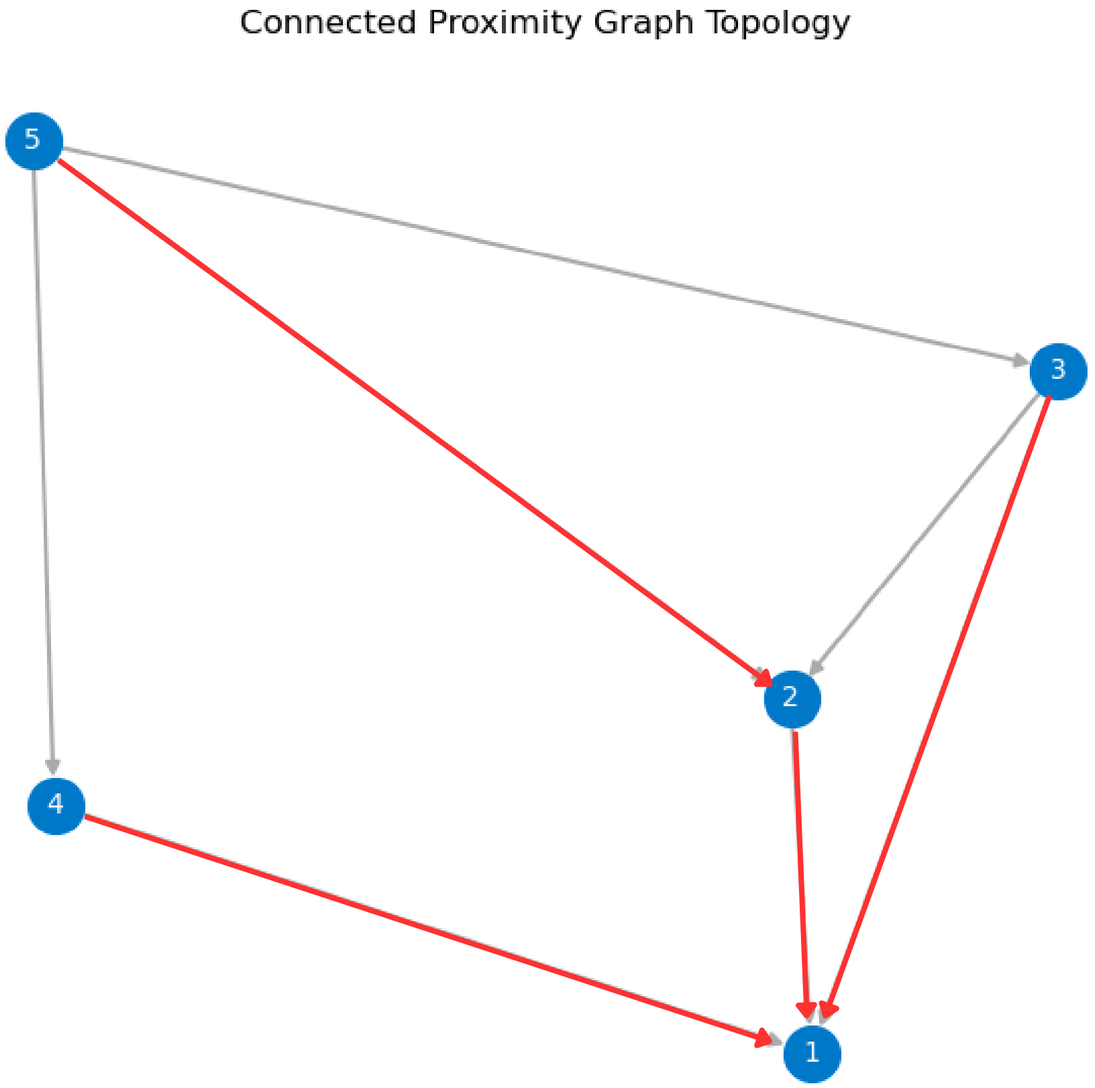
- BlueRov2 model no input constraints
- [Research challenge] integrate input constraints
- [Research challenge] controller robust or less sensitive to uncertainties (e.g. inverse optimal controller in the sense of LgV like control laws rather than feedback linearization)

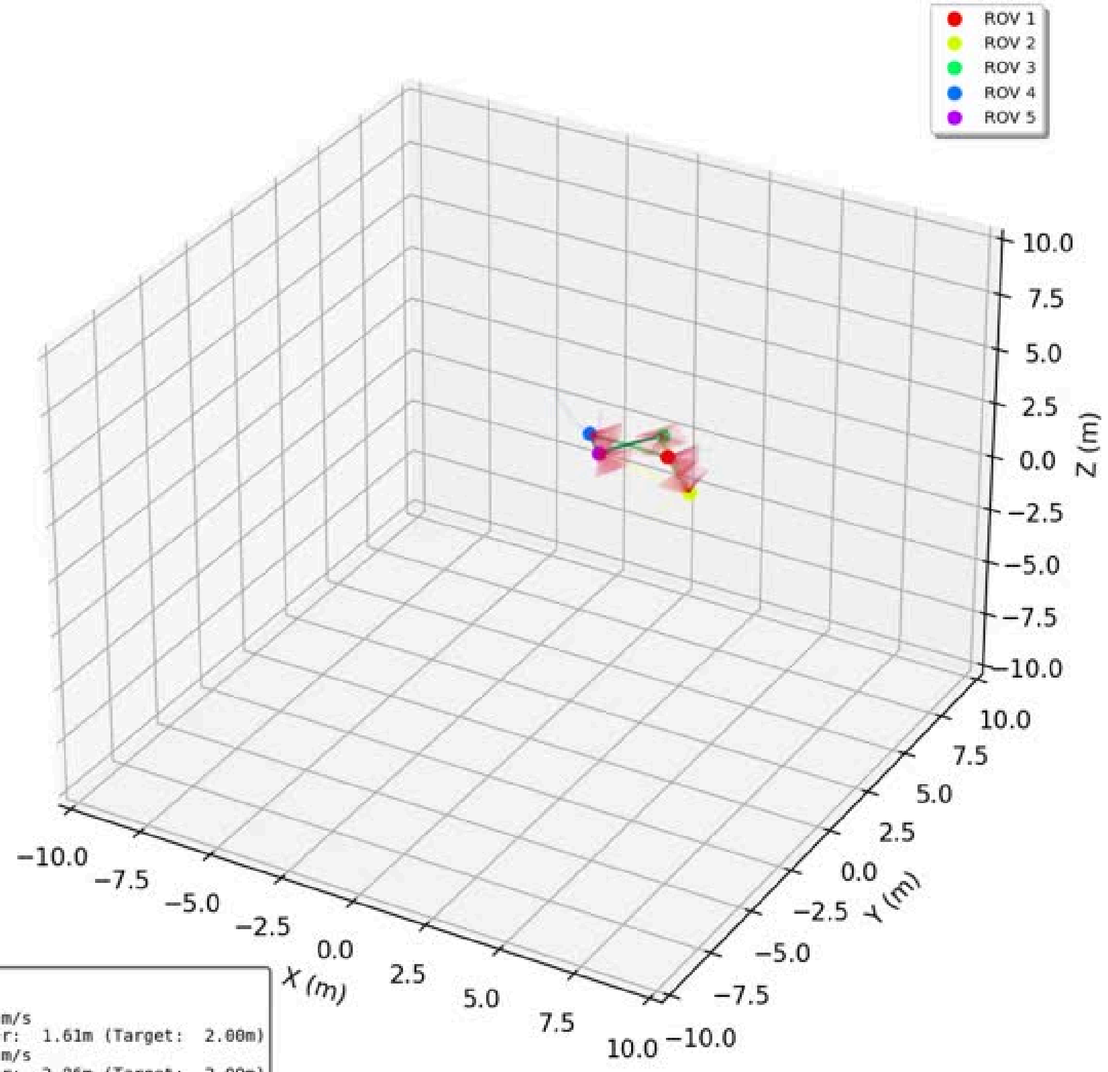
Achievements

- Study
  - Graph Theoretic Methods in Multiagent Networks
  - Esteban’s thesis
  - Khalil's book on Nonlinear Control

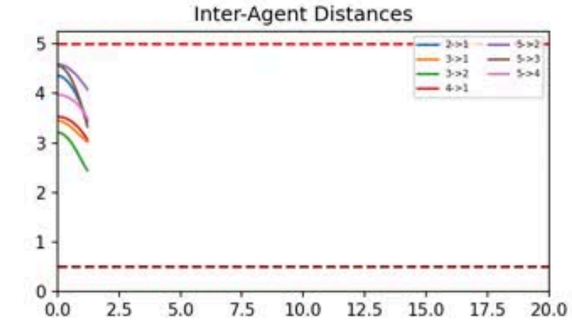
- Implementation
  - Improvements
    - Field of view error plot (horizontal and vertical)
    - Stablished target agent regarding the fov
    - Dynamic graph edges on simulation
    - Recentered barrier function
- Wrote Report

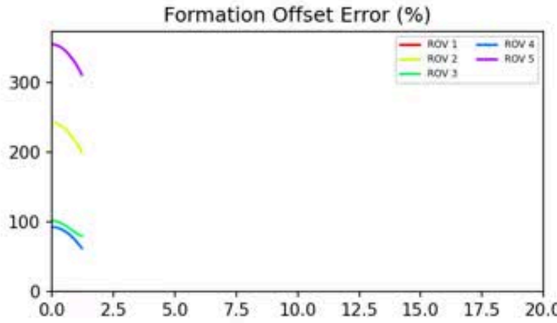

Connected Proximity Graph Topology
5
3
2
4
1

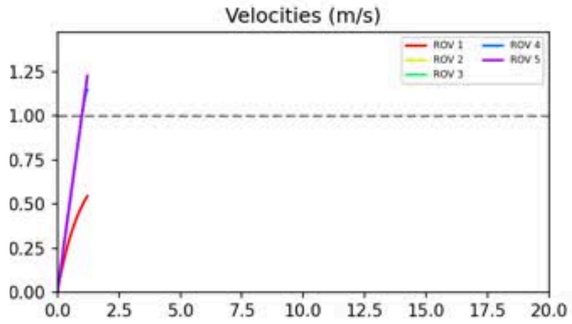

Connected Proximity Graph Topology
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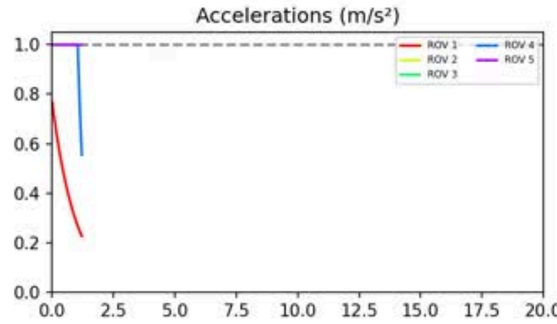


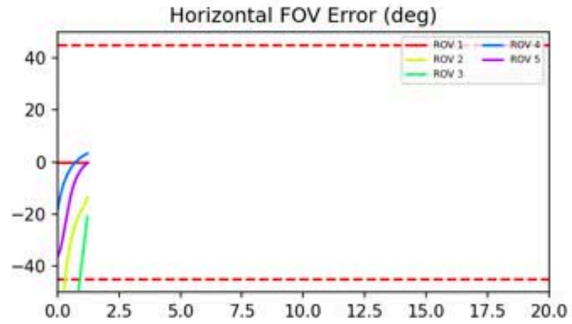

```
Time:  2.36s
---------------
ROV 1 Vel:  0.70 m/s
Distance to Leader:  1.61m (Target:  2.00m)
ROV 2 Vel:  2.04 m/s
Distance to Leader:  2.86m (Target:  2.00m)
ROV 3 Vel:  0.46 m/s
Distance to Leader:  2.56m (Target:  2.00m)
ROV 4 Vel:  0.79 m/s
Distance to Leader:  3.11m (Target:  2.00m)
ROV 5 Vel:  2.38 m/s
```

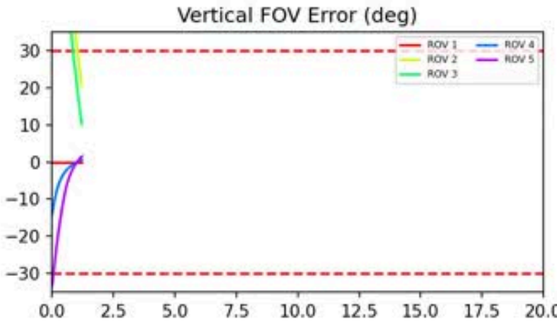


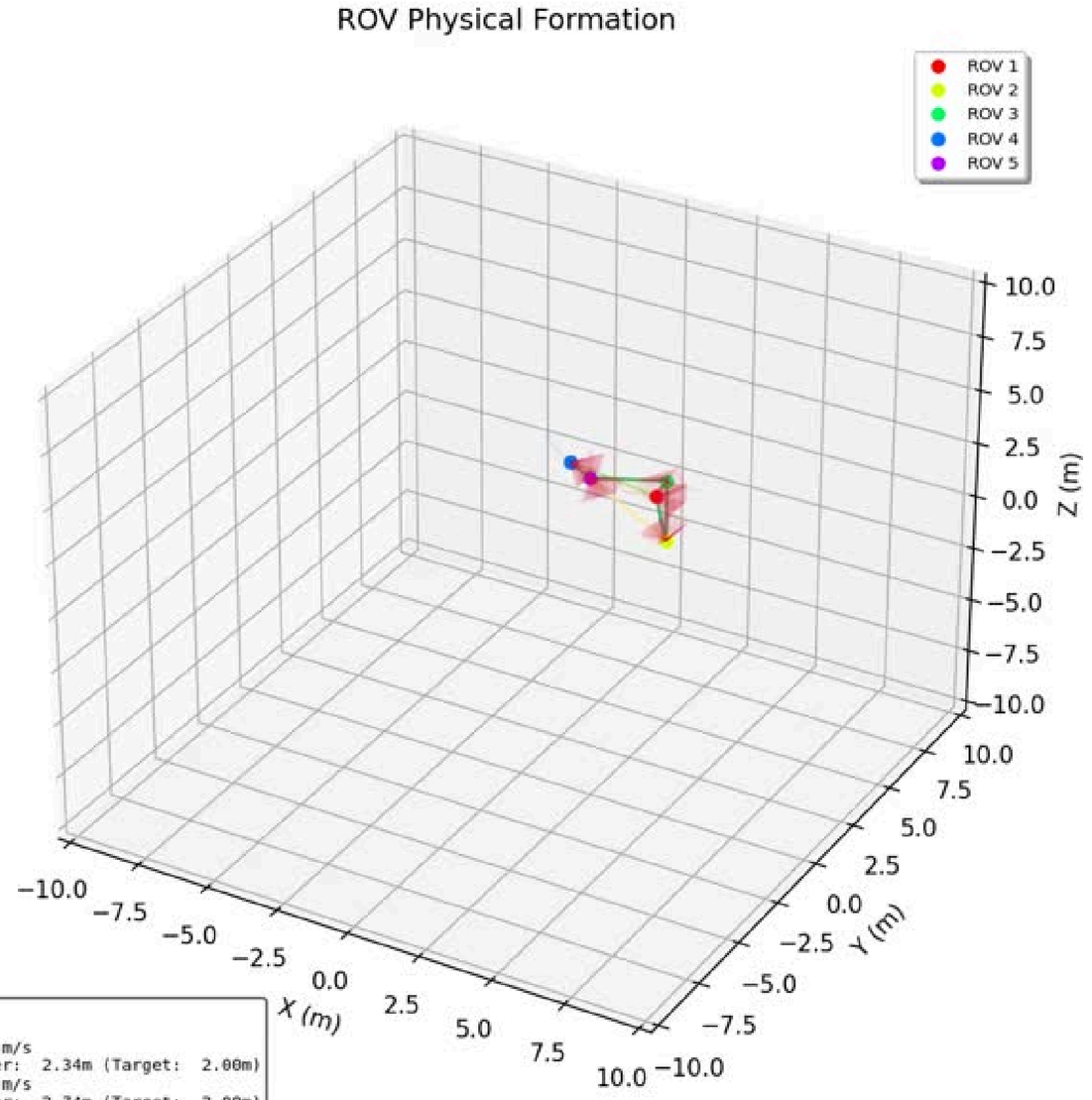




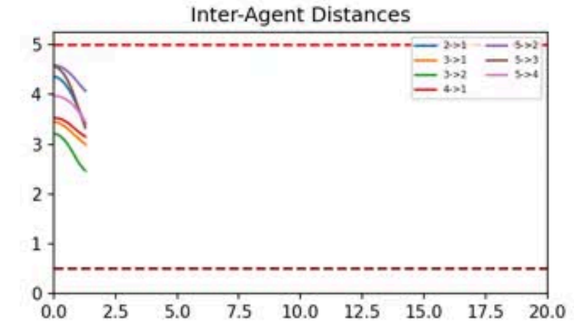


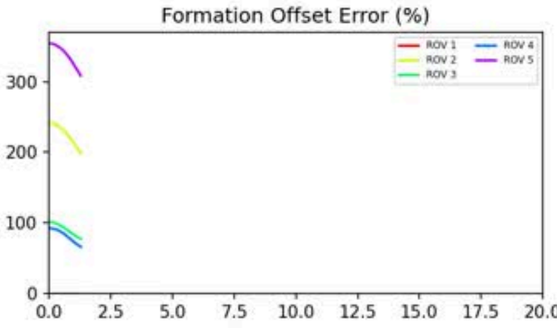


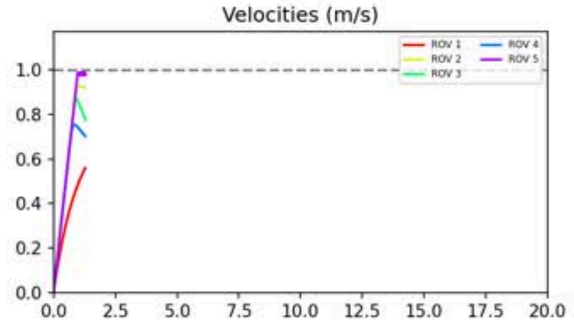


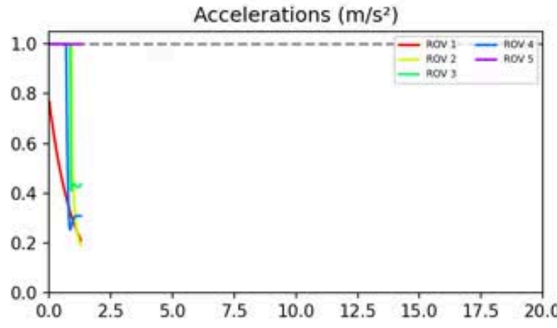


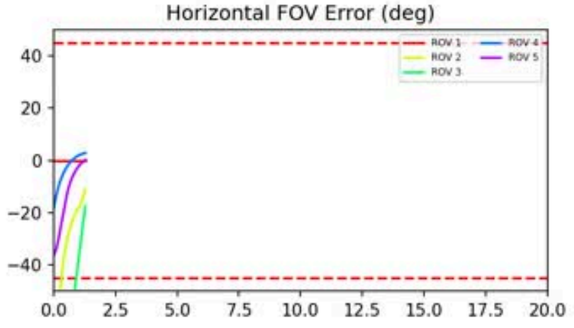

```
Time:  2.32s
---------------
ROV 1 Vel:  0.69 m/s
Distance to Leader:  2.34m (Target:  2.00m)
ROV 2 Vel:  0.88 m/s
Distance to Leader:  2.74m (Target:  2.00m)
ROV 3 Vel:  0.20 m/s
Distance to Leader:  2.97m (Target:  2.00m)
ROV 4 Vel:  0.64 m/s
Distance to Leader:  3.84m (Target:  2.00m)
ROV 5 Vel:  0.97 m/s
```

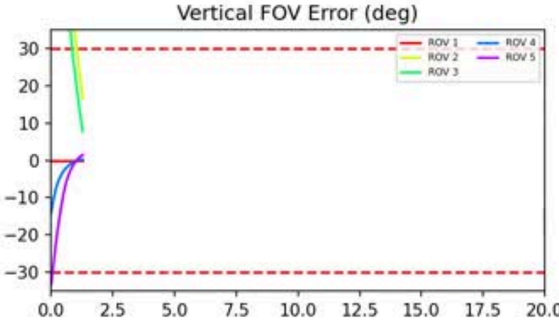












Recentered Barrier Function

$$v^* := -c_1[E_\odot \otimes I_3]\nabla W(\tilde{z})$$

$$W_k(\tilde{z}_k) = \frac{1}{2}\left[|\tilde{z}_k|^2 + \tilde{B}_k(\tilde{z}_k + z_k^d)\right],$$

$$\begin{aligned}\tilde{B}_k(\tilde{z}_k + z_k^d) =& \kappa_{1,k}\left[\ln\left(\frac{\Delta_k^2}{\Delta_k^2 - |\tilde{z}_k + z_k^d|^2}\right) - \ln\left(\frac{\Delta_k^2}{\Delta_k^2 - |\tilde{z}_k + z_k^d|^2}\right)\right] \\ &+ \kappa_{2,k}\left[\ln\left(\frac{|\tilde{z}_k + z_k^d|^2}{|\tilde{z}_k + z_k^d|^2 - \delta_k^2}\right) - \ln\left(\frac{|z_k^d|^2}{|z_k^d|^2 - \delta_k^2}\right)\right],\end{aligned}$$

$$\kappa_{1,k} := \frac{\delta_k^2}{|z_k^d|^2(|z_k^d|^2 - \delta_k^2)}, \quad \kappa_{2,k} := \frac{1}{\Delta_k^2 - |z_k^d|^2}.$$

$$\nabla\bar{B}(0) = \left[\left(\frac{1}{\Delta_k^2 - |z_0|^2}\right) - \left(\frac{1}{|z_0|^2 - \delta_k^2}\right)\right] z_0$$

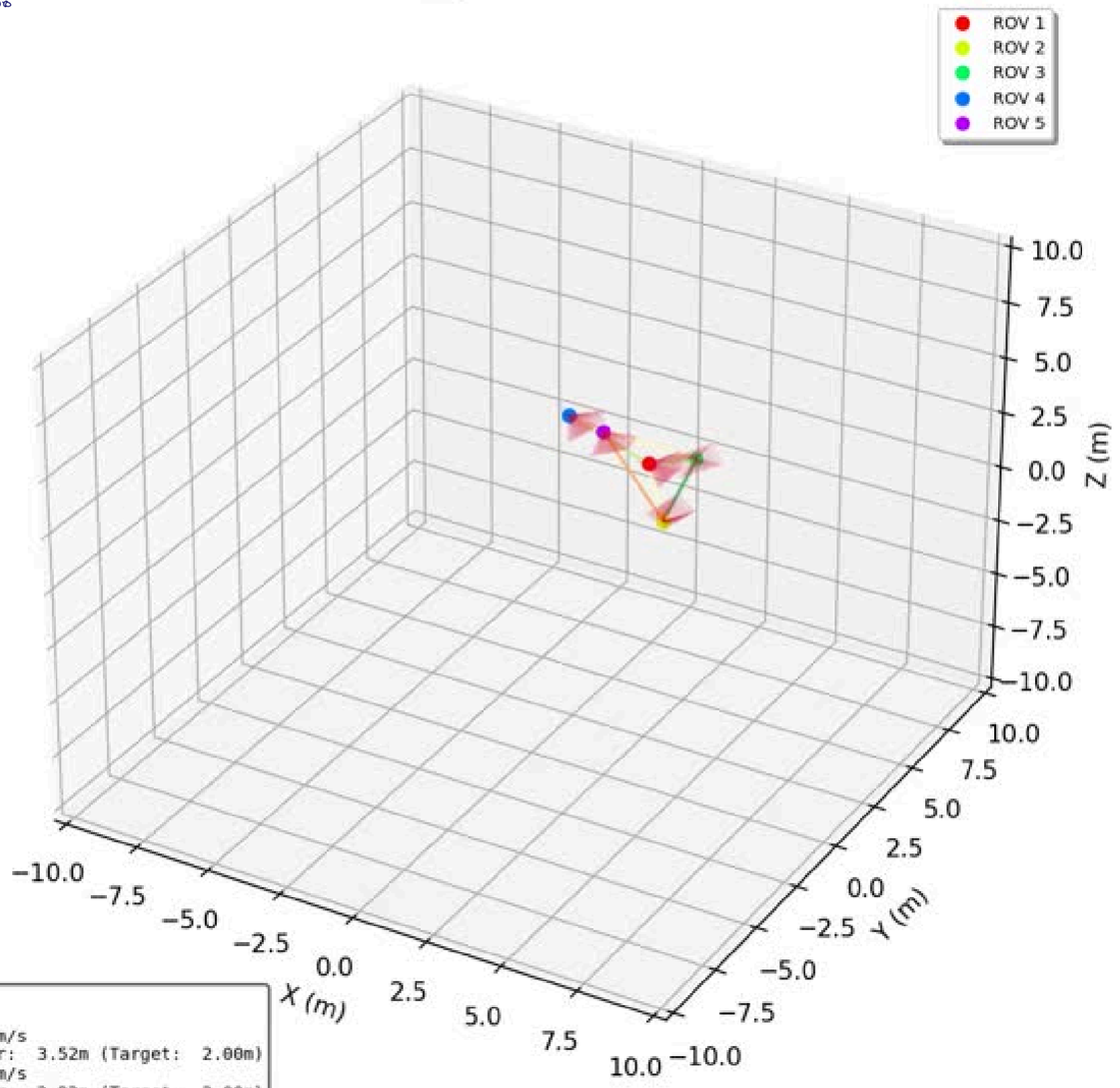
$$\nabla\bar{B}(z) = \left[\left(\frac{1}{\Delta_k^2 - |z_k|^2}\right) - \left(\frac{1}{|z_k|^2 - \delta_k^2}\right)\right] z_k$$

Definition 1 (Gradient recentered barrier function [9]) *Let $\bar{B} : D \to \mathbb{R}$ be a convex barrier function on an open convex set D with $0 \in D$. Then, the function $B : D \to \mathbb{R}$ with*

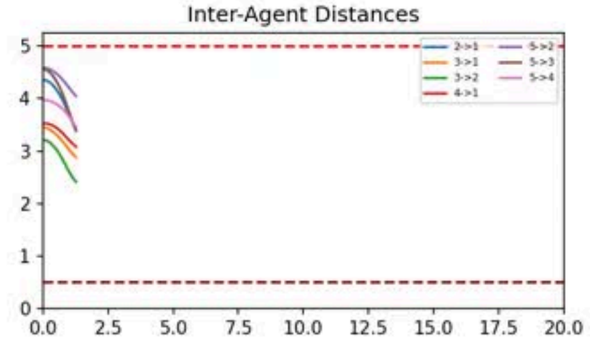
$$B(z) = \bar{B}(z) - \bar{B}(0) - [\nabla\bar{B}(0)]^\top z \tag{4}$$

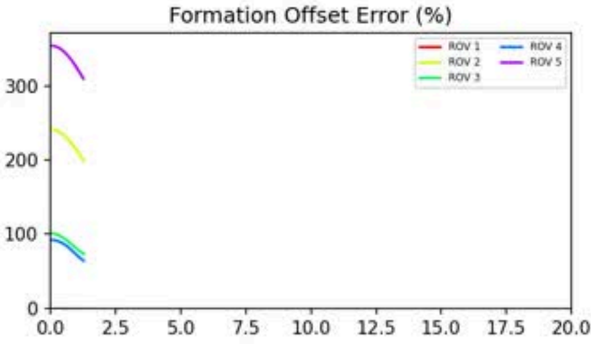
defines the gradient recentered barrier function for $\bar{B}$ around the origin. It clearly holds that $B(0) = 0$, $\nabla B(0) = 0$.

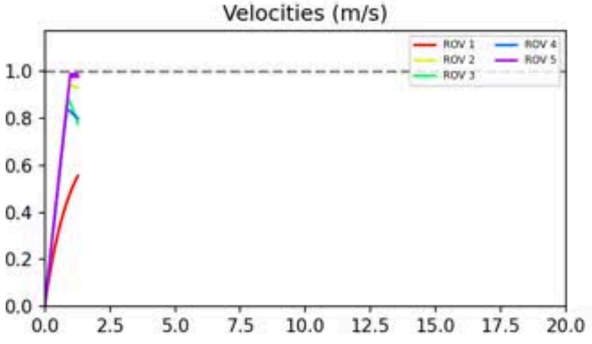
$$\begin{aligned}\nabla B(z) &= \nabla\left(\bar{B}(z) - \bar{B}(0) - \left[\nabla\bar{B}(0)\right]^T z\right) \\ &= \nabla\bar{B}(z) - \underbrace{\nabla\bar{B}(0)}_{0} - \nabla\left[\nabla\bar{B}(0)\right]^T z \\ &= \nabla\bar{B}(z) - \nabla\bar{B}(0)\end{aligned}$$

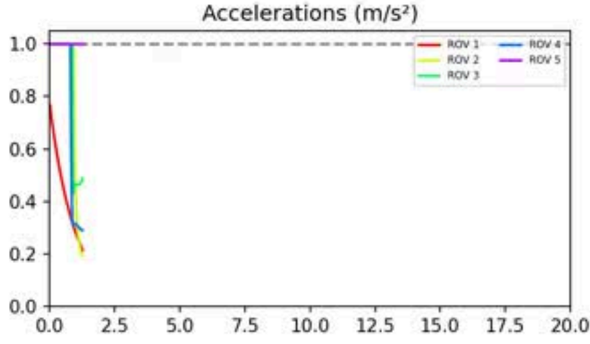


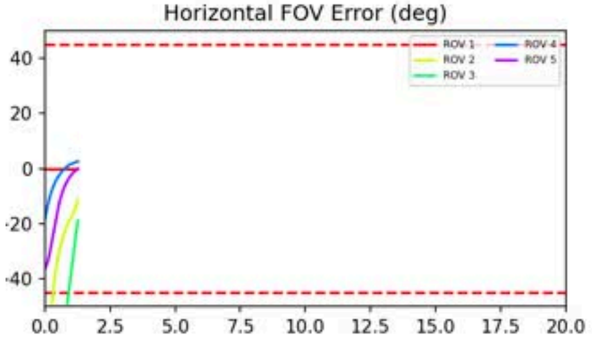

```
me:  1.16s
-------------
V 1 Vel:  0.53 m/s
stance to Leader:  3.52m (Target:  2.00m)
V 2 Vel:  0.93 m/s
stance to Leader:  2.93m (Target:  2.00m)
V 3 Vel:  0.81 m/s
stance to Leader:  3.13m (Target:  2.00m)
V 4 Vel:  0.81 m/s
stance to Leader:  4.83m (Target:  2.00m)
V 5 Vel:  0.98 m/s
```

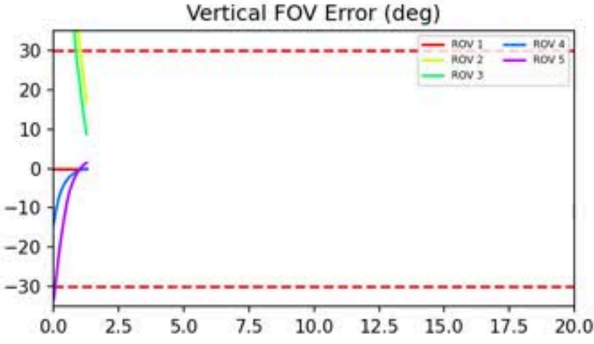


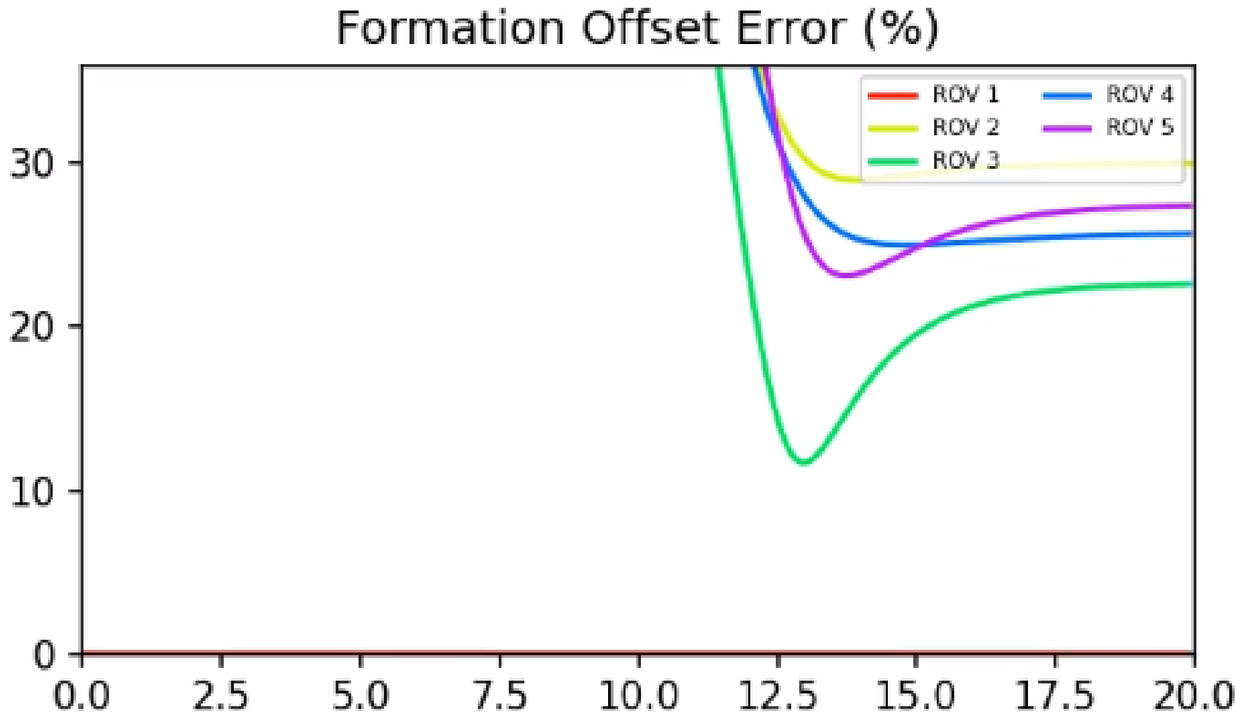


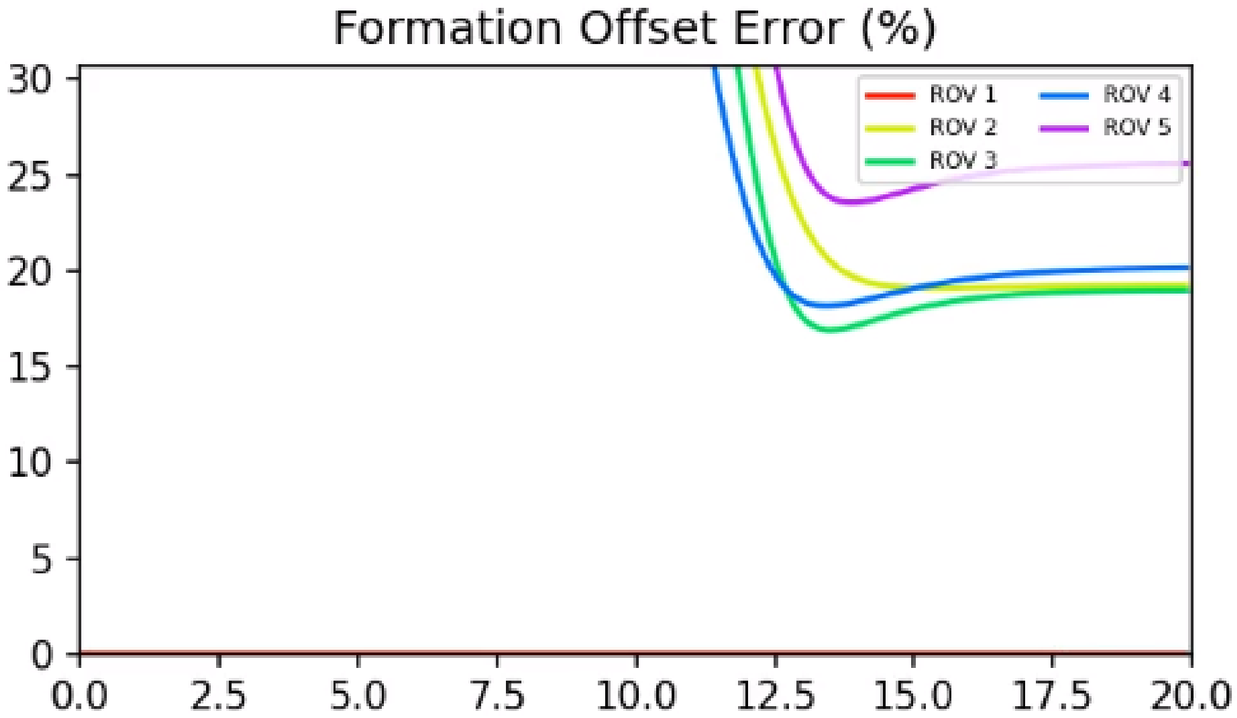









Formation Offset Error (%)
ROV 1
ROV 2
ROV 3
ROV 4
ROV 5
30
20
10
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Formation Offset Error (%)
ROV 1
ROV 2
ROV 3
ROV 4
ROV 5
30
25
20
15
10
5
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

Next Steps

- Formation control with single integrator agents no constraints
  - Undirected graph
  - Directed graph
- Single integrator formation control with constraints
  - Only connectivity maintenance
  - Then also colision avoidance
  - Field of view
- Double integrator agents

- BlueRov2 model no input constraints
- [Research challenge] integrate input constraints
- [Research challenge] controller robust or less sensitive to uncertainties (e.g. inverse optimal controller in the sense of LgV like control laws rather than feedback linearization)



Questions & Comments